Paleomagntism of Archean rocks from northwestern Ontario: II. Shelley Lake granite, Quetico Subprovince

DAVID J. DUNLOP
Geophysics Laboratory, Department of Physics, University of Toronto, Toronto, Ont., Canada M5S 1A7

Received December 2, 1983
Revision accepted March 8, 1984

The 2580 ± 20 Ma Shelley Lake granite of the Quetico gneiss belt in northwestern Ontario preserves two distinct natural remanent magnetizations (NRM's) of different ages. Type 1 NRM of either normal (1N) or reverse (1R) polarity was isolated in 43 samples, using three methods: stable end-point directions, orthogonal vector plots, and converging remagnetization circles. The mean direction based on both 1N and 1R results is $D = 5.9°$ $I = +56.4°$ ($k = 95.4$, $\alpha_{95} = 4.4°$, $N = 12$ sites), which is significantly different from both the present Earth's field (PEF) and axial dipole field directions at the site. Type 2 NRM, either normal (2N) or reverse (2R), was successfully separated from 1N/1R in 13 samples. Its mean direction, combining both polarities, is $D = 70.3°$ $I = -27.8°$ ($k = 20.1$, $\alpha_{95} = 9.4°$, $N = 13$ samples).

The paleopole SL1 corresponding to 1N/1R falls at 65.8°E, 77.6°N, near track 6 of the Laurentian apparent polar wander path (APWP) around 2600 Ma. This remanence is carried principally by primary multidomain magnetite with blocking temperatures up to 580°C. If it is a primary thermoremanence, as is argued here, SL1 provides a well dated tie point for the APWP at 2580 ± 20 Ma.

Paleopole SL2 corresponding to 2N/2R falls at 157.1°W, 1.3°S, around 1250 Ma on the APWP. The remanence is probably a chemical or thermochemical overprint acquired during a mild heating event ($T < 300$°C, $t > 1100$ Ma) that reset $^{40}Ar/^{39}Ar$ feldspar ages. An earlier tentative assignment of SL2 to track 6 around 2800 Ma is incorrect.

Le granite du lac Shelley daté à 2580 ± 20 Ma appartient à la ceinture des gneiss de Quetico du nord-ouest de l'Ontario et comporte deux aimantations rémanentes naturelles (ARN) d'âges différents. L'ARN de type 1, de polarité normale (1N) et (ou) inversée (1R), fut retracé dans 43 échantillons en utilisant trois méthodes: point final stable des directions, courbes du vecteur orthogonal et cercles convergents de réaimantation. La direction moyenne déduite des résultats de 1N et 1R correspond à $D = 5{,}9°$ $I = +56{,}4°$ ($k = 95{,}4$, $\alpha_{95} = 4{,}4°$, $N = 12$ emplacements), laquelle divergent considérablement des directions actuelles du champ terrestre et du dipôle axial de l'emplacement. L'ARN du type 2, de polarité normale (2N) ou inversée (2R), fut aisément reconnue comme étant différente de 1N/1R dans 13 échantillons. Sa direction moyenne, les deux polarités étant combinées, est $D = 70{,}3°$ $I = -27{,}8°$ ($k = 20{,}1$, $\alpha_{95} = 9{,}4°$, $N = 13$ échantillons).

Le paléopole SL1 qui correspond à 1N/1R pointe à 65,8°E, 77,6°N, près du tracé 6 de la courbe de la dérive apparente des pôles (CDAP) laurentienne, approximativement 2600 Ma. Cette rémanence est portée principalement par les domaines multiples de la magnétite primaire dont les températures de blocage peuvent atteindre jusqu'à 580°C. Si la thermorémanence est primaire, comme nous le supposons ici, SL1 représente alors une date fiable pour la CDAP à 2580 ± 20 Ma.

Le paléopole SL2 correspondant à 2N/2R pointe à 157,1°O, 1,3°S, environ 1250 Ma sur la CDAP. Cette rémanence résulte probablement d'une surimpression chimique ou thermochimique acquise lor d'un événement thermique peu élevé ($T < 300$°C, $t > 1100$ Ma) qui a provoqué la remise à zéro des âges des feldspaths déterminés par $^{40}Ar/^{30}Ar$. L'assignation proposée de SL2 au tracé 6, environ 2800 Ma, énoncée antérieurement se révèle maintenant incorrecte.

[Traduit par le journal]

Can. J. Earth Sci. **21**, 869–878 (1984)

Introduction

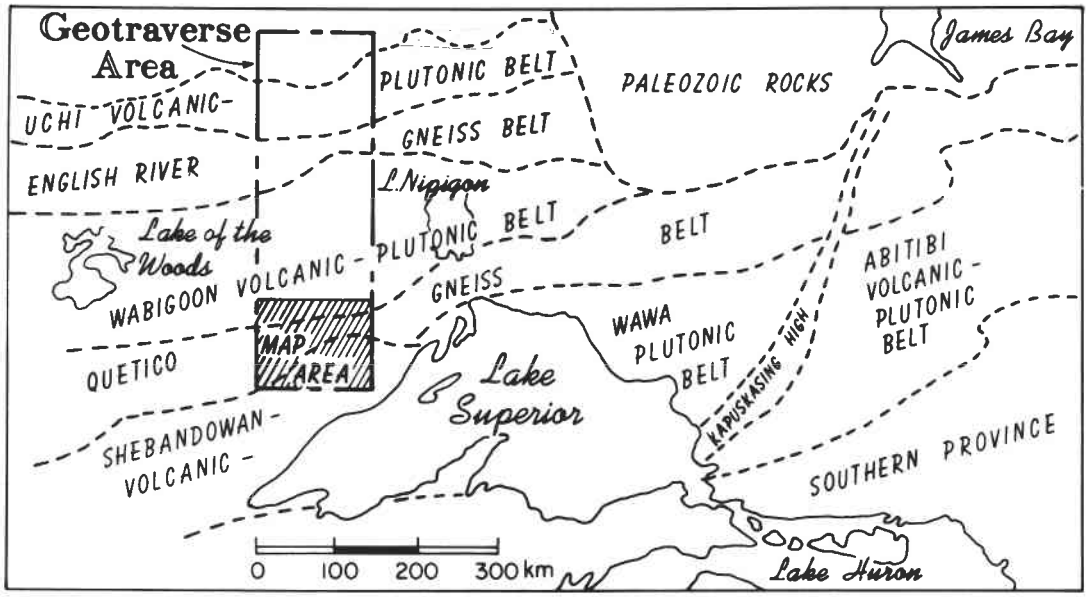
Archean paleopoles for North America are few in number and for the most part poorly dated. Apart from early work on the Stillwater Complex of Wyoming (Bergh 1970), most studies have concentrated on low-grade (prehnite–pumpellyite to greenschist) rocks of the Superior Province Abitibi belt (Fig. 1*a*) (Irving and Naldrett 1977; Schutts and Dunlop 1981; Symons *et al.* 1981; Geissman *et al.* 1982, 1983; Tasillo-Hirt *et al.* 1982). In the Abitibi region, there are a number of examples of late Archean magnetizations that escaped total overprinting in the ≈2600 Ma Kenoran Orogeny. The late Archean apparent polar wander path (APWP) as presently known relies heavily on these results.

Recently an apparently pre-Kenoran magnetization has been detected in the Wabigoon gabbro of northwestern Ontario (Dunlop 1983), confirming preliminary results of a regional survey of the "Superior Geotraverse" corridor (Fig. 1*a*) (Dunlop 1979*a*). The Wabigoon gabbro is an undated greenschist-grade intrusion in the Superior Province Wabigoon belt. The present paper investigates the granitic Shelley Lake pluton, which intrudes amphibolite-grade schists of the more southerly Quetico gneiss belt. The Shelley Lake granite is confidently dated at 2580 ± 20 Ma by $^{40}Ar/^{39}Ar$ (Berger and York 1979), and provides a well dated Kenoran reference paleopole anchoring the late Archean APWP.

In addition to the pervasive magnetization believed to date from initial cooling around 2580 Ma, some Shelley Lake samples are overprinted by a remanence that superficially resembles the pre-Kenoran magnetization found in the Wabigoon and Abitibi belts but that actually dates from late Proterozoic time. Its age, by association with updated $^{40}Ar/^{39}Ar$ microcline ages, is >1100 Ma (Berger and York 1979). It probably formed in a mild (<300°C) thermal event associated with Keweenawan rifting in the nearby Lake Superior basin.

Geological Setting

The Shelley Lake granite intrudes Archean schists of the Quetico gneiss belt in northwestern Ontario (Fig. 1*b*). Modal mineral abundances in Shelley Lake thin sections average 68.5% microcline and orthoclase, 3.5% albite, 24.5% quartz, 3% biotite (altered to varying degrees to chlorite), trace amounts of muscovite and sphene, and up to 1% opaques, mainly magnetite. As a result of the high concentration of magnetite, the Shelley Lake body is marked by a distinct aero-



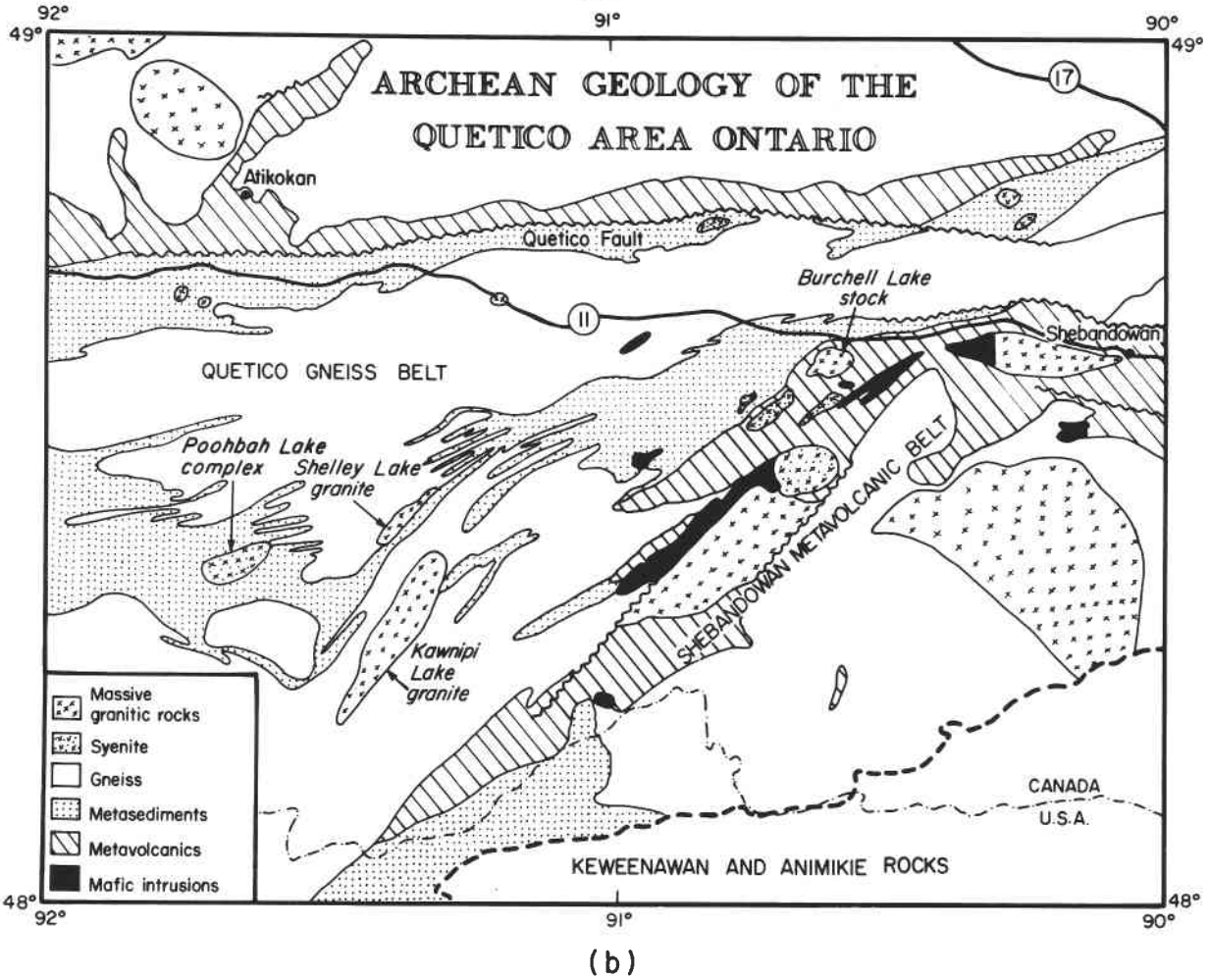


FIG. 1. (*a*) Subprovinces of the Superior Province in central and western Ontario, showing the locations of the Superior Geotraverse area and the map area of (*b*). (*b*) Regional geology of the Quetico area, indicating the location of the Shelley Lake granite. Also shown are the Poohbah Lake alkaline complex and the Burchell Lake stock, to be described in future papers in this series.

magnetic anomaly. In fact, the aeromagnetic expression is the only feature distinguishing the Shelley Lake from surrounding lithologically similar orthogneisses.

The Shelley Lake intrusion is post-kinematic. It is fresh and undeformed compared with the sheared amphibolite-grade schists it intrudes. $^{40}Ar/^{39}Ar$ plateau ages of 2560 and 2595 Ma for hornblende and biotite, respectively (Berger and York 1979; G. W. Berger, personal communication, 1980), are conclusive evidence of rapid cooling between about 600 and 300°C, the respective argon blocking temperatures of these two minerals. The Shelley Lake pluton must have been emplaced into country rock that was already comparatively cool at 2580 ± 20 Ma.

Sampling and measurement procedures

Sixty-four oriented block samples were collected from 12 sites within the Shelley Lake aeromagnetic high. Although the samples are unusually magnetic for granites, with natural remanent magnetizations (NRM's) generally $< 2 \times 10^{-3}$ emu cm^{-3} (2 A m^{-1}), there was no difficulty in orienting samples using a magnetic compass. No contact test was attempted, since the only accessible contact with Couchiching series schists to the southeast forms the channel of the Maligne River.

In the laboratory, three or four 2.5 cm specimens were drilled from each block. A full program of measurements was carried out on every sample. One specimen was thermally demagnetized, usually in 50°C steps to 500°C and then in 25°C steps to 675°C, using a noninductive furnace in a three-axis Helmholtz-coil array. The remaining two or three specimens were alternating field (AF) demagnetized in 12 steps to 1000 Oe (0.1 T) in a Schonstedt demagnetizer. Magnetizations were measured after each step with a Digico spinner magnetometer.

Results were accepted if specimen magnetization vectors achieved a stable grouping different from the present Earth's field (PEF) at higher AF's or temperatures and if within-sample divergences were <20°. The AF/temperature level that minimized within-sample dispersion of stable end points or subtracted vectors was used in sample, site, and formation averages. For the converging remagnetization circle technique, all NRM vector swings that followed great-circle paths were accepted.

Thermomagnetic curves in a field of 3000−5000 Oe (0.3−0.5 T) were measured for 22 samples in air or in vacuum with a recording Curie balance (Bailey 1980). Optical microscopy at magnifications up to 1000× was carried out by L. D. Schutts.

Nature of the magnetic carriers

Curie point and thin section observations will be considered in detail in a companion paper (Dunlop *et al.* 1984, referred to as III in the rest of this paper). A summary is given here.

Thin sections revealed four types of magnetic opaques. In order of decreasing abundance, they are: (1) primary euhedral to subhedral magnetite crystals 50−500 μm in diameter, slightly to moderately subdivided by lamellae in {111} planes; (2) hematite (not ilmenite) lamellae, probably late-stage deuteric; (3) widespread post-crystallization hematite, staining fractures and edges of primary opaques; and (4) scarce fine-grained secondary opaques, probably magnetite, a by-product of chloritization of biotites.

Electron microprobe analyses demonstrated that Ti is not present in either the groundmass or lamellae of the large magnetite crystals. Vacuum thermomagnetic curves were essentially reversible and indicated only pure magnetite, with a Curie point near 580°C, and hematite.

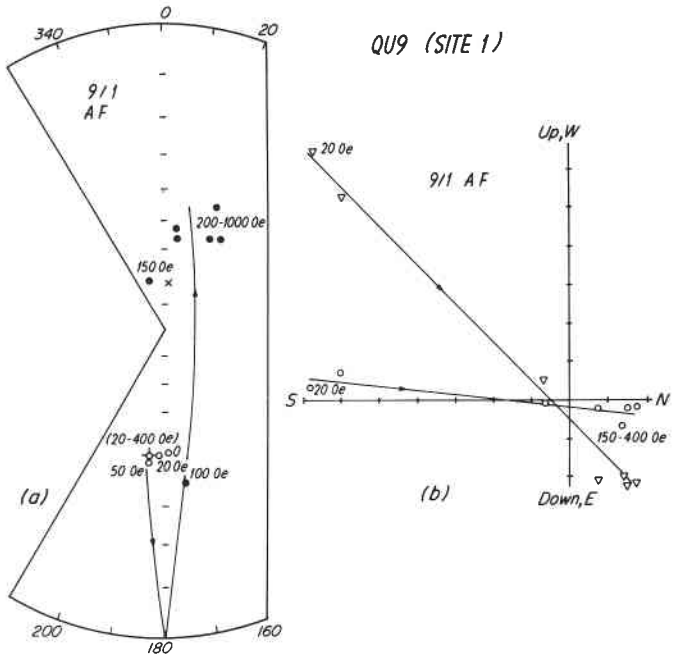


FIG. 2. Behaviour of the NRM vector of one specimen of sample QU9 in the course of AF demagnetization. (*a*) Stereographic projection of magnetization unit vectors. Solid and open symbols in this and succeeding figures denote downward and upward vectors, respectively. The PEF is indicated by a cross. Symbols marked by a vertical cross and labelled in parentheses (e.g., 20−400 Oe) indicate subtracted vectors over the specified AF or temperature range. (*b*) Orthogonal vector plot; circles denote horizontal-plane projections and triangles denote vertical-plane (Up vs. N or Up vs. E) projections.

Koenigsberger Q_n ratios for about one third of the collection were <0.5, reflecting truly multidomain (MD) behaviour of the coarse primary opaques. The remaining samples had Q_n's ranging from 0.5 to 7, a hallmark of single-domain (SD) or pseudo-SD carriers. Lowrie and Fuller (1971) tests gave mainly MD results.

NRM results, type 1

Typical demagnetization characteristics

Two thirds of the Shelley Lake samples have a stable type 1 magnetization like that of QU9 in Fig. 2. The NRM vector is either southward and up (1R) or northward and down (1N), the polarity being arbitrarily defined with respect to the PEF. QU9 has a composite NRM in which 1R and 1N are superimposed. 1R, the predominant component, resides in soft MD magnetite (see paper III) with coercivities (H_C's) mostly between 50 and 150 Oe (5 and 15 mT) (Fig. 2*b*) and blocking temperatures (T_B's) up to 580°C. The smaller 1N component is very nearly 180° reversed to 1R (Fig. 2*b*) and resides in hematite with $H_C > 1000$ Oe (0.1 T) and $T_B > 580$°C. The 1N direction tends to be 10−20° shallower than the PEF (Fig. 2*a*).

Sample QU48 (Fig. 3) behaves in much the same manner as QU9, except that a complete reversal to 1N was not achieved during cleaning of most specimens. 1R has some H_C's above 400 Oe (40 mT) in specimen 48/1 and most of its H_C's exceed 300 Oe (30 mT) in 48/2. 1R is again stable in thermal cleaning to 580°C. Only at 624 and 659°C does a second component begin to be revealed. Judging by the stable end point reached

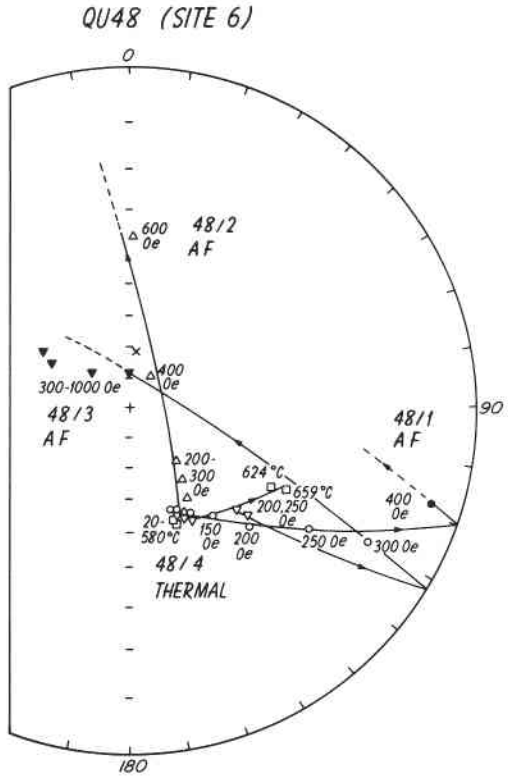


FIG. 3. Stereoplot of demagnetization behaviour of sample QU48, whose dominant NRM is 1R.

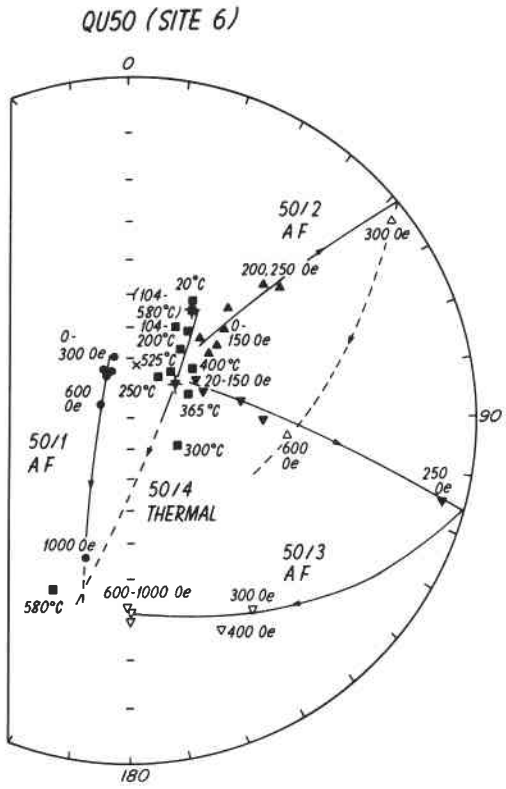


FIG. 4. Stereoplot of AF and thermal demagnetization vector trajectories for sample QU50, in whose NRM 1N and 1R vectors are superimposed.

by 48/3, the second component is probably 1N, but the NRM intensity dropped below the instrumental noise level before it could be isolated. Although no 1N end point was reached in three of the four specimens, the fact that great-circle swings of the NRM vector have a broad azimuthal spread implies 180° reversals.

In some samples, of which QU50 (Fig. 4) is an example, the roles of 1R and 1N are interchanged: 1N is carried by MD or pseudo-SD magnetite with $T_B < 580°C$ and 1R resides in hematite. AF cleaning is again of varying efficiency in isolating the high-H_C or high-T_B vector. At 600 Oe (60 mT), the 1R end point is reached in specimen 50/3, approached in 50/2, but not revealed at all in specimen 50/1. Such variability within a hand sample is typical of the collection. It reflects variability in the relative strengths of 1N and 1R components and, to a lesser extent, variability in the coercivity spectra of the mineral carriers.

In summary, type 1 NRM's of both normal and reverse polarities frequently coexist in individual samples. One polarity is carried by low-H_C MD or pseudo-SD magnetite with blocking temperatures up to 580°C and the other polarity is carried by high-H_C, high-T_B hematite. Self-reversal is a possibility, given the intimate association of hematite lamellae and primary magnetite.

However, at least one field reversal must have occurred while the type 1 remanence was being acquired, since the low-temperature remanence is in some instances 1R and in others 1N.

Mean directions from stable end points and subtracted vectors

A three-tier statistical analysis of type 1 data is given in Table 1. Both end-point directions and subtracted vectors determined from linear segments in orthogonal vector plots

TABLE 1. Shelley Lake granite, type 1 site-mean directions after AF or thermal cleaning

Site	N	D (°)	I (°)	k	α_{95} (°)
1	4	4	67.5	17	23
2	6	23	50	32	12
3	4	11	52.5	11	29
4	4	20	65	42	19
5	4	357	58	12	28
6	4	2	52	18	22
7	4	14	48	16	24
8	5	4	51	13	22
9	3	353	52	17	31
10	1	3	64		
11	3	355	56	8	46
12	1	4	57		
Mean, unit weight to samples	43	7.5	55.4	17.4	5.4
Mean, reverse samples only	14	186.0	−53.9	11.9	12.4
Mean, normal samples only	29	8.2	56.1	20.8	5.9
Mean, unit weight to sites	12	5.9	56.4	95.4	4.4
Paleopole		65.8E (δp=4.7°, δm=6.4°)	77.6N		

NOTES: N is the number of samples or sites averaged; D, I are declination and inclination of magnetization; k is Fisher's precision parameter; α_{95} is the radius of the cone of 95% confidence about the mean direction; and δp, δm are semi-axes of the 95% confidence oval around the paleopole.

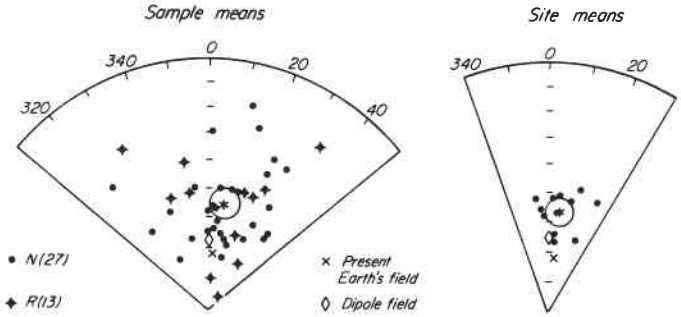


FIG. 5. Equal-area projections of AF and thermally cleaned sample-mean and site-mean type 1 magnetizations for the Shelley Lake granite. The grand means are indicated by stars and are surrounded by their circles of 95% confidence.

(Zijderveld 1967; Dunlop 1979*b*) are included. Samples not included were weakly or unstably magnetized, heavily overprinted by the PEF, or struck by lightning. These results were discarded. A small number of samples exhibited type 2 NRM, to be considered in the next section.

To allow for the possibility that some PEF overprints might have been mistaken for 1N remanences, separate averages were calculated for 1R and 1N data. The 1R data are somewhat more scattered than the 1N data, but the averages are indistinguishable: $D = 8.2°$ $I = +56.1°$ for 1N; $D = 186.0°$ $I = -53.9°$ for 1R. Thus within the uncertainty limits of the data, the reversals are 180°.

Sample-mean and site-mean averages are similarly indistinguishable (Fig. 5), although the site-mean precision ($k = 95.4$) is considerably higher than the sample-mean precision ($k = 17.4$). Circles of 95% confidence for sample-mean and site-mean averages exclude both the PEF and the axial dipole field directions for a latitude of 48°N. On this evidence, as well as the existence of reverse polarities and the comparatively high blocking temperatures of both 1N and 1R, the type 1 remanence is ancient.

Mean directions from intersecting remagnetization circles

The numerous vector swings that do not reach stable end points can be used to estimate mean 1N and 1R directions by projecting the incomplete great-circle trajectories to a common point of intersection. This is the intersecting remagnetization circle technique of Halls (1976). The wide spread in the azimuths at which great circles cross the horizontal plane makes for good definition of the intersection point.

Figure 6*a* illustrates some typical trajectories intersecting near 1R. Each great circle is the trace of the remagnetization plane defined by the resultant of 1N and 1R vectors as one or the other is progressively erased during demagnetization. In Fig. 6*b*, each such plane is represented by its vector normal (in this diagram, the upward normals have been chosen). The best fitting great circle through all the normals has been calculated using the program of Bailey and Halls (1984), adapted for microcomputer use by C. Jackson. In turn, the normal to the plane represented by this great circle is the best estimate of the intersection point of the original demagnetization trajectories.

A statistical analysis is given in Table 2. The mean intersection point is $D = 169.7°$ $I = -55.4°$, with a precision $k' = 35.5$ (corresponding to the Fisher precision k of stable endpoint data) and semi-axes of the confidence oval $\beta_{95} = 6.0°$ and $\gamma_{95} = 8.2°$ for $N = 28$ specimen normals. This estimate of the 1R vector compares well with the grand mean of 1R stable end

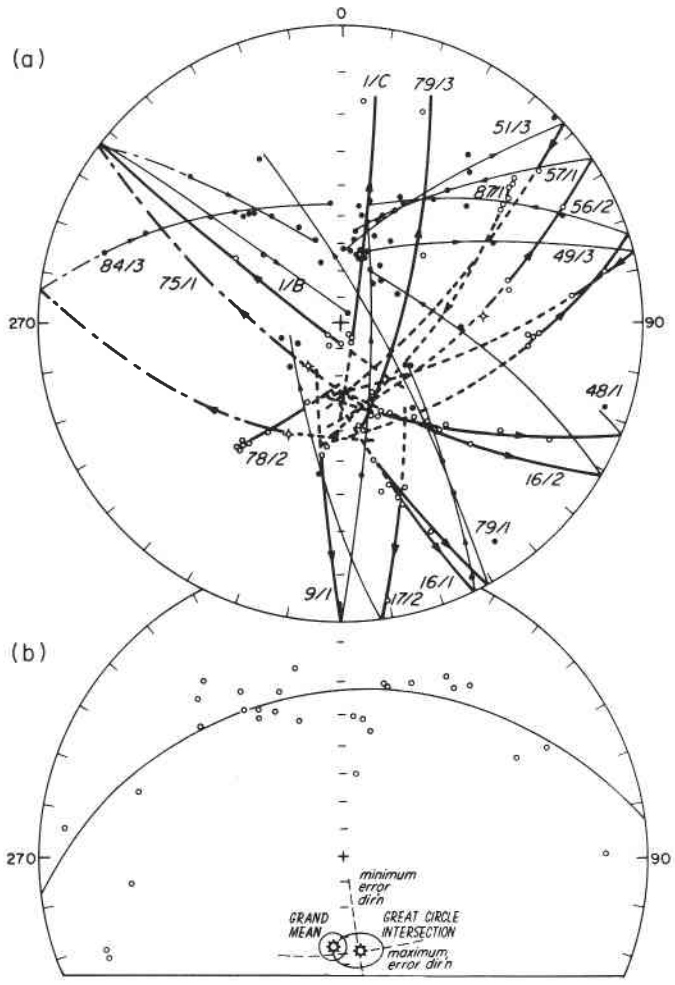


FIG. 6. (*a*) Stereographic projection of typical demagnetization trajectories of NRM vectors in the course of AF or thermal cleaning. The trajectories shown here intersect near the mean 1R direction. (*b*) Stereoplot of the normals to remagnetization planes whose traces on the unit sphere are the great-circle trajectories in (*a*), and the best fitting great circle through the normals. The normal to this great circle is the best estimate of the intersection point of the original trajectories. The great-circle intersection point agrees well with the 1R grand mean direction.

points (Table 1). The confidence oval around the mean intersection point (Fig. 6*b*) takes into account least-squares deviations of remagnetization normals from the best fitting great circle (Bailey and Halls 1984). It is analogous to the circle of 95% confidence about the mean of a Fisherian population of unit vectors.

The great-circle intersection analysis is repeated in Fig. 7 and Table 2 for lower-hemisphere demagnetization trajectories intersecting near the 1N direction. The mean intersection point, $D = 0.3°$ $I = +60.5°$, is similar to the 1N mean and of comparable precision ($k' = 74.4$, $\beta_{95} = 3.8°$, $\gamma_{95} = 6.4°$).

The northeast–southwest elongations of the confidence ovals reflect a relative abundance of great circles with east to northeast or west to southwest azimuths, dictated in most cases by the presence of type 2 NRM vectors, with either northeast or southwest declinations and shallow inclinations.

NRM results, type 2

About a dozen Shelley Lake samples possess type 2 NRM's. Both polarities are present, with "normal" (east to northeast

TABLE 2. Shelley Lake granite, type 1 directions from remagnetization planes intersecting near 1N or 1R

	N	D (°)	I (°)	k'	β_{95} (°)	γ_{95} (°)
IR, unit weight to specimen normals	28	169.7	−55.4	35.5	6.0	8.2
Paleopole		123.7E	75.4N			
		(δp<8.6°, δm<11.7°)				
1N, unit weight to specimen normals	29	0.3	60.5	74.4	3.8	6.4
Paleopole		6.7E	83.0N			
		(δp<7.7°, δm<9.8°)				

NOTES: Statistical parameters k', β_{95}, γ_{95} are explained in the text. Other symbols as in Table 1.

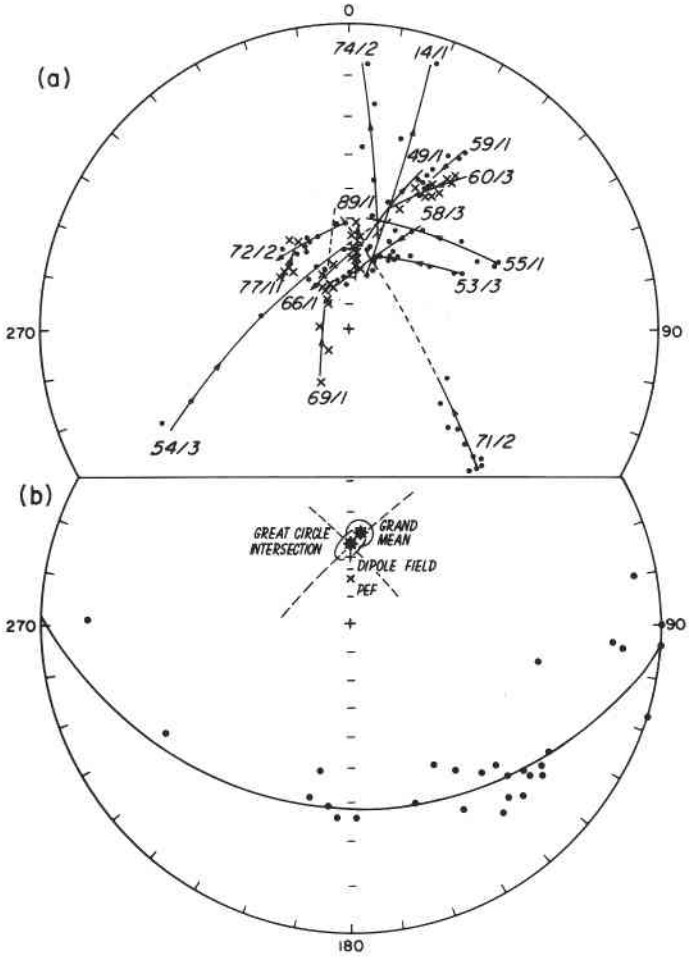


FIG. 7. Demagnetization trajectories intersecting near 1N and the best estimate of the intersection point, which agrees well with the 1N + 1R grand mean.

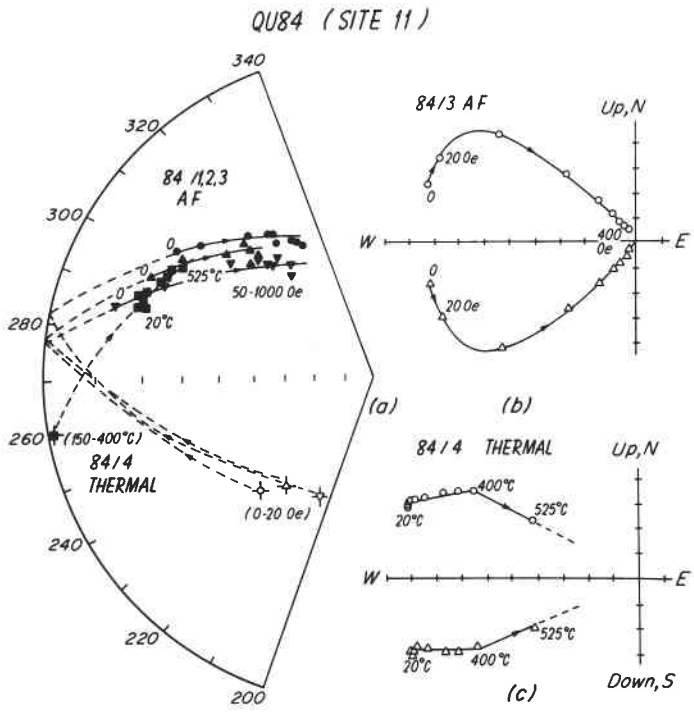


FIG. 8. 1R + 2R + 1N multivectors in sample QU84. 2R is resolved in thermal cleaning (ideal vector plot in (*c*)) but not in AF cleaning (non-ideal) vector plot in (*b*)).

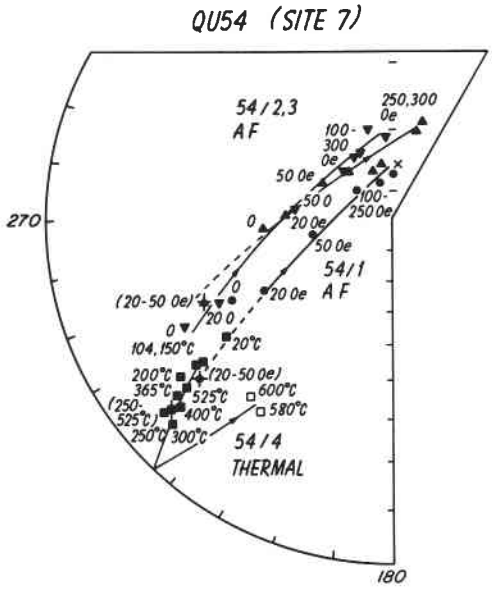


FIG. 9. Stereoplot of 2R → 1N AF swings and a 2R → 1R thermal swing in sample QU54.

declination) being more frequent. 2N and 2R vectors are not usually isolated as stable end points by thermal or AF cleaning. They tend to have low or intermediate stability and must be separated from coexisting 1N and 1R NRM's by means of vector diagrams.

A strongly preferred azimuth for all specimen demagnetization trajectories from a given sample is a good indication that a type 2 remanence is present. This is the case for the examples illustrated in Figs. 8−10 and contrasts with the randomly oriented trajectories seen in Figs. 3, 4, and 6. The 2N or 2R vector in effect "pins" the otherwise arbitrary 1N−1R remagnetization plane.

Three NRM vectors are revealed in the stereoplot and vector diagrams of QU84 (Fig. 8). Between 150 and 400°C, a 2R component near $D = 260°$ $I = 0°$ is removed and the resultant NRM vector migrates towards a 1N direction. On AF demagnetization, all specimens reveal a large, soft 1R component and a <180°-reversed, hard 1N component. The AF vector diagram (Fig. 8*b*) is non-ideal; the presence of a 2R remanence is probably masked because its coercivities overlap those of 1R and 1N.

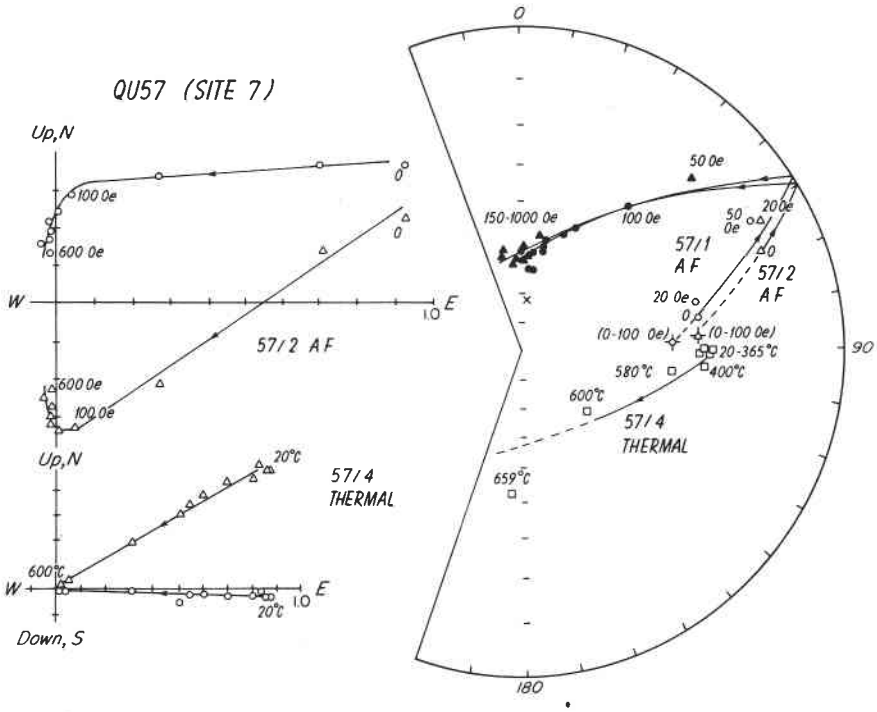


FIG. 10. 2N + 1N (AF) and 2N + 1R (thermal) behaviour in sample QU57.

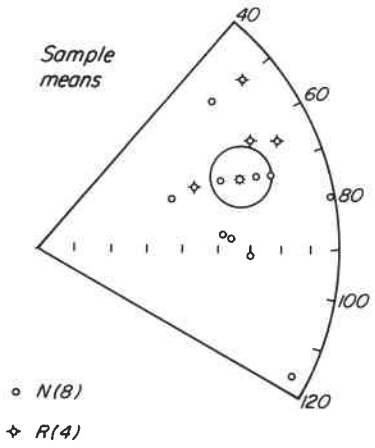


FIG. 11. Equal-area projection of AF and thermally cleaned sample-mean type 2 magnetizations.

In the previous example, a 2R vector was prominent in thermal treatment but not revealed in AF cleaning. In QU54 (Fig. 9), on the other hand, a soft 2R component is erased between 20 and 50 Oe (2 and 5 mT) in specimens 54/1 and 3 and a hard 1N component is isolated at higher AF's. On thermal demagnetization, after elimination of what is probably a PEF overprint (20−250°C), a 2R component is removed between 250 and 525°C and the residual vector migrates towards 1R at higher temperatures. QU79 (not shown) has a similar behaviour.

QU57 in Fig. 10 and QU56 (not illustrated) behave in a manner reminiscent of QU54, except that 2N rather than 2R is the third vector. The AF vector plot is non-ideal, but 2N is clearly a very soft magnetization and 1N a comparatively hard one. On thermal cleaning, 2N is the predominant remanence. Only above 580°C is a very small 1R component revealed.

In summary, type 2 NRM occurs sporadically at 8 of the 12

TABLE 3. Shelley Lake granite, type 2 mean direction after AF or thermal cleaning

	N	D (°)	I (°)	k	α_{95} (°)
Mean, unit weight to samples	13	70.3	−27.8	20.1	9.4
Paleopole		157.1W	1.3S		
		(δp=6.1°, δm=10.4°)			

NOTE: Symbols as in Table 1.

Shelley Lake sites. It is usually a multivectorial composite with 1R and 1N remanences. The mineral phase or phases carrying type 2 remanence tend to have soft or intermediate coercivities overlapping those of the soft MD magnetite and hard hematite that are the principal sources of 1N and 1R vectors. 2N and 2R blocking temperatures are usually >250°C but less than those of coexisting 1R or 1N remanences. 2N/2R is therefore probably younger than 1N/1R. It must be a chemical rather than a thermal overprint, since its maximum laboratory T_B is quite variable and since it occurs haphazardly, even at the site level.

Sample-mean type 2 vectors, both stable end points and subtracted vectors, are plotted in Fig. 11. A two-tier statistical analysis appears in Table 3. The mean type 2 direction is D = 70.3° I = −27.8° (k = 20.1, α_{95} = 9.4°, N = 13 samples).

Paleopoles and apparent polar wander

Type 1

The paleomagnetic pole SL1 corresponding to the grand mean type 1 direction lies at 65.8°E, 77.6°N (Table 1), with δp = 4.7°, δm = 6.4°. Its position is near the late Archean APWP for Laurentia (Fig. 12) for an assumed magnetization age of 2500−2600 Ma.

This is an entirely reasonable age since the Shelley Lake pluton initially cooled around 2580 Ma and the 1N/1R remanence is carried in part by primary magnetite and in part by

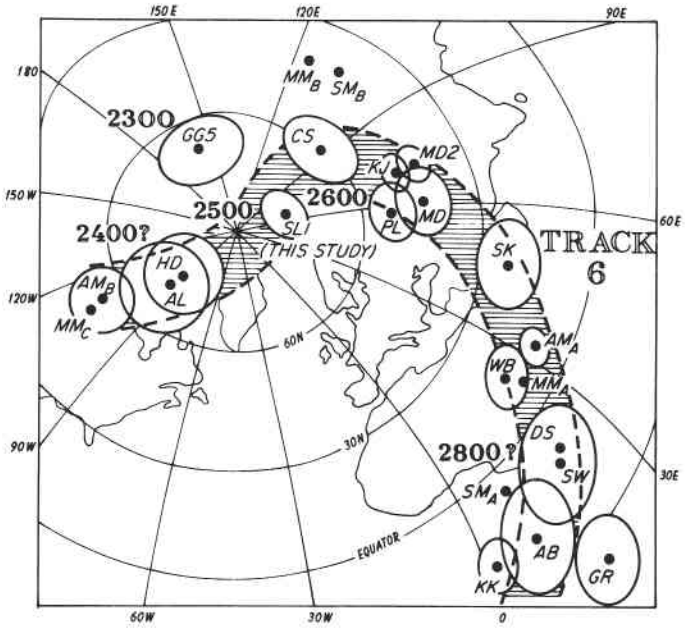


FIG. 12. Tentative late Archean APWP for Laurentia, following Irving (1979, Fig. 11) and Irving and McGlynn (1981, Fig. 23–14), with additional primary and secondary poles for a number of Archean iron formations after Symons *et al.* (1981) and Seguin *et al.* (1983). Other poles include AB and MD2 (Abitibi belt country rocks and Matachewan dikes, Schutts and Dunlop 1981), PL (Poohbah Lake complex, Quetico belt, Dunlop 1979*a*), SK (Skead Group pyroclastics, Abitibi belt, Ridler and Foster 1981), GR and KJ (Ghost Range Complex and Kinojevis volcanics, Abitibi belt, Geissman *et al.* 1982, 1983), WB (Wabigoon gabbro, Wabigoon belt, Dunlop 1983), and SL1 of the present study. The position of SL1 relative to the APWP is consistent with its proposed primary magnetization age of 2580 ± 20 Ma.

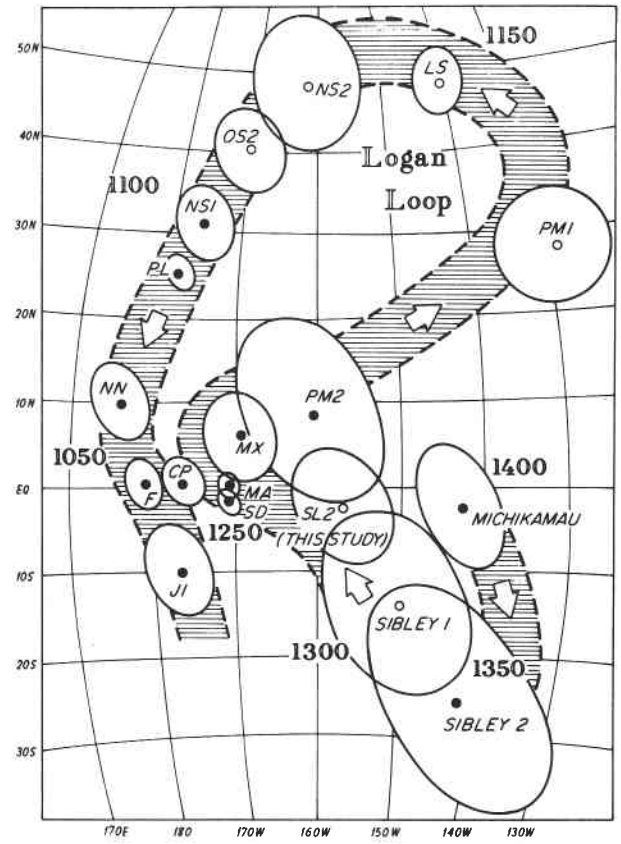


FIG. 13. Late Proterozoic APWP for Laurentia, after Halls and Pesonen (1982, Fig. 4*a*), to which most of the labels are keyed, with the addition of Mackenzie-age (1250 Ma) poles MX, CP, MA, and SD and the Michikamau anorthosite (1400 Ma) pole from Irving (1979, Fig. 9). Pole SL2 of the present study falls on the APWP near 1250 Ma, but it is probably related to Keweenawan rather than Mackenzie heating and remagnetization.

hematite. There are both deuteric and post-crystallization hematites in these samples, but it is plausible that if the primary magnetite grains acquired a thermoremanent magnetization (TRM) in initial cooling, the hematite lamellae cutting the magnetite should have acquired the same (actually a reverse) TRM. Secondary hematite may play a role also, but if so, the exact 180° separation of 1N and 1R vectors implies that any chemical remanent magnetization (CRM) from this source was produced soon after cooling before significant apparent polar wander occurred. Alternatively, the CRM might have been directionally controlled by the primary TRM.

Type 1 NRM is therefore argued to be primary and to record the paleofield about 2580 Ma ago. The Quetico Subprovince was at that time located about 10° farther from the geographic pole than it now is but was in very nearly its present orientation.

Type 2

The paleopole SL2 corresponding to the mean type 2 direction falls at either 22.9°E, 1.3°N, or 157.1°W, 1.3°S (Table 3), with $\delta p = 6.1°$, $\delta m = 10.4°$. The former position is close to 2700–2800 Ma paleopoles for the Wabigoon gabbro, a number of Abitibi formations, and the Stillwater Complex (uncorrected for structure). However, the group 2 NRM cannot be as old as 2700 Ma because the Shelley Lake granite first cooled around 2580 Ma and bears a primary magnetic imprint of this age.

In a preliminary report of data from the Shelley Lake and a number of other Superior Geotraverse formations (Dunlop 1979*a*), a pre-Kenoran age for the group 2 paleopole was considered a real possibility. At that time, the $^{40}Ar/^{39}Ar$ biotite age of 2595 Ma for the Shelley Lake granite had been determined (Berger and York 1979) but the hornblende age of 2560 Ma, which was decisive in demonstrating rapid cooling from high temperatures, had not yet been measured. Irving (1979, Fig. 11) incorporated the preliminary group 2 pole, which he labelled SA, in his Archean track 6 of the Laurentian APWP. SA has since been cited by other authors (Irving and McGlynn 1981; Symons *et al.* 1981; Geissman *et al.* 1982, 1983; Tasillo-Hirt *et al.* 1982; Seguin *et al.* 1983) but the 2700 Ma age assignment is clearly wrong and SA should be deleted from track 6.

Instead SL2, as the type 2 pole is more appropriately designated, lies on the upper Proterozoic track 4A of the APWP (Fig. 13) just before its junction with track 3 (the Logan Loop). It is bracketed by a number of poles associated with the Mackenzie igneous episode (CP, MX, SD, MA) and by the 1300 Ma pole S1 of the uppermost Sibley redbeds, which underlie the Keweenawan rift volcanic sequence in the Lake Superior basin. A mild heating event in the Quetico area associated with either Mackenzie–Sudbury dike intrusion or the early stages of Keweenawan rifting would account for the type 2 overprint.

That such an event occurred is not in doubt. $^{40}Ar/^{39}Ar$ step-heating analyses of two microcline separates by Berger and York (1979) demonstrated that the Shelley Lake body experienced a mild thermal pulse (150–300°C maximum tem-

perature) lasting 1 Ma or so with a minimum age of 1100 Ma. This age is suggestive of Keweenawan overprinting rather than Mackenzie (although it should be noted that 1100 Ma poles do *not* resemble SL2) and the proximity of the Keweenawan rift to the southeast makes this association a virtual certainty. There are no known Mackenzie-age dikes within several hundred kilometres of the Quetico area, and a search for Mackenzie-age activity in the Wabigoon Subprovince (Dunlop 1983) was fruitless.

To summarize, type 2 NRM is argued to be a low-temperature chemical or thermochemical overprint acquired during a brief thermal pulse about 1250 Ma ago, likely associated with incipient Keweenawan rifting. At this time, the Quetico Subprovince was near the equator and rotated 90°, present-day west being paleonorth.

Discussion

The Shelley Lake granite probably records reversals of the Earth's magnetic field around 1250 Ma (2N/2R) and 2580 Ma (1N/1R). For the type 2 remanence, the evidence is not compelling, since 2N and 2R are CRM's. One can imagine a situation in which 2N, for instance, formed parallel to the ambient field and at some later time 2R formed antiparallel to 2N. But there is no doubt that the field reversed during initial cooling of the Shelley Lake pluton. 1N and 1R are present in different samples as primary TRM's with identical blocking temperature ranges of 250−580°C.

The 1N/1R field reversal was essentially 180°. 1R and 1N means are indistinguishable (Table 1) and the azimuths of 1N/1R great circles are widely dispersed. Kenoran reversals differ in this respect from Keweenawan reversals, which some authors have proposed may be asymmetric (e.g., Halls and Pesonen 1982).

The type 1 paleopole (SL1) falls distinctly north of track 6 as conventionally drawn (Irving 1979; Irving and McGlynn 1981). Berger and York (1979) pointed out, however, that there is an apparent hiatus in continental drift between ≈2600 Ma (pole CS in Fig. 12) and ≈2300 Ma (pole GG5) according to track 6. Such a dramatic cessation of drift is unlikely.

It seems more sensible to tentatively adopt an APWP based on the one proposed by Symons *et al.* (1981) to incorporate primary and secondary magnetizations from a number of Ontario Archean iron formations. This is the 2600−2400 Ma path depicted in Fig. 12. It links in a natural sequence poles MD and MD2 of the 2630 ± 90 Ma Matachewan diabase (Irving and Naldrett 1977; Schutts and Dunlop 1981), CS of the 2570 ± 170 Ma Chibougamau sills (Ueno and Irving 1976), SL1 at 2580 ± 20 Ma, and AL and HD of Algoman granite and Haileyburian diorite with estimated magnetization ages of ≈2500 Ma (Londry and Symons 1976). MD and SL1 are two of the best dated Archean paleopoles anchoring the Laurentian APWP.

Lapointe *et al.* (1978) have suggested that high-latitude poles are underrepresented in the paleomagnetic data set. SL1, at 66°E, 78°N, is such a pole (although the implied paleolatitude of 36°N is 12° south of the present latitude of the Shelley Lake area). Moreover SL1 plays a pivotal role in that it is being proposed as a tie point for the Laurentian APWP. How certain can one be that the type 1 remanence is not a PEF overprint? There are three lines of evidence: (1) "reversed" polarities (i.e., opposed to the PEF) occur frequently; (2) that part of the type 1 NRM carried by magnetite exhibits blocking temperatures up to 580°C, much in excess of T_B's associated with viscous overprinting at ordinary temperatures (Wilson and Smith 1968; Dunlop 1969; Pullaiah *et al.* 1975); and (3) the circle of 95% confidence about the type 1 mean excludes both the PEF and the axial dipole field.

The paleomagnetic interpretation of SL1 as a primary TRM hinges on rock magnetic observations described in paper III. The key pieces of evidence are the observation of very large primary crystals of magnetite as the principal opaque mineral, the characterization of these grains as MD, and the matching of their synthetic AF decay curves with the soft AF characteristics of 1N/1R remanence. It appears that granites, because of their unusual and distinctive magnetic carriers, which combine high blocking temperatures with low coercivities, make ideal subjects for the interplay between rock magnetism and paleomagnetism.

Conclusions

(1) The dominant magnetization of the Shelley Lake granite is a 1N/1R remanence, believed to be in large part a primary TRM, with an age of 2580 ± 20 Ma and a direction $D = 5.9°$ $I = 56.4°$ ($k = 95.4$, $\alpha_{95} = 4.4°$, $N = 12$ sites). The corresponding paleopole is 65.8°E, 77.6°N ($\delta p = 4.7°$, $\delta m = 6.4°$).

(2) The average intersection point of remagnetization circles was indistinguishable in direction and precision from the grand mean of stable end points and subtracted vectors.

(3) The geomagnetic field reversed 180° during cooling of the Shelley Lake granite.

(4) The Shelley Lake granite is sporadically overprinted by a 2N/2R remanence that originated as a CRM or TCRM sometime in the interval 1300−1100 Ma, most likely around 1250 Ma. Incipient Keweenawan rifting rather than Mackenzie igneous activity was the likely agent for mild regional heating and remagnetization. The mean 2N direction is $D = 70.3°$ $I = -27.8°$ ($k = 20.1$, $\alpha_{95} = 9.4°$, $N = 13$ samples) with a paleopole at 157.1°W, 1.3°S ($\delta p = 6.1°$, $\delta m = 10.4°$).

Acknowledgments

Many people contributed to the success of this project, but above all I thank Drs. Kenneth Buchan and Monika Bailey, who shared many companionable seasons in the field, and Christine Blackburn for her devotion to the measurements and data reduction. Srinivas Thaneer and Mark Stirling assisted both in the field and the laboratory. The work was supported by an Energy, Mines and Resources Canada research agreement and by Natural Sciences and Engineering Research Council of Canada operating and equipment grants.

BAILEY, M. E. 1980. Magnetic properties of deep-ocean basalts. Ph.D. thesis, University of Toronto, Toronto, Ont., 121 p.

BAILEY, R. C., and HALLS, H. C. 1984. Estimate of confidence in paleomagnetic directions derived from mixed remagnetization circle and direct observational data. Journal of Geophysics. (In press.)

BERGER, G. W., and YORK, D. 1979. $^{40}Ar-^{39}Ar$ dating of multicomponent magnetizations in the Archean Shelley Lake granite, northwestern Ontario. Canadian Journal of Earth Sciences, **16**, pp. 1933−1941.

BERGH, H. W. 1970. Paleomagnetism of the Stillwater Complex, Montana. *In* Paleogeophysics. *Edited by* S. K. Runcorn. Academic Press, New York, NY, pp. 143−158.

DUNLOP, D. J. 1969. Comments on a paper by R. L. Wilson and P. J. Smith: The nature of secondary magnetizations in some igneous and baked rocks. Journal of Geomagnetism and Geoelectricity, **21**,

pp. 797–799.

——— 1979*a*. A regional paleomagnetic study of Archean rocks from the Superior Geotraverse area, northwestern Ontario. Canadian Journal of Earth Sciences, **16**, pp. 1906–1919.

——— 1979*b*. On the use of Zijderveld vector diagrams in multicomponent paleomagnetic studies. Physics of the Earth and Planetary Interiors, **20**, pp. 12–24.

——— 1983. Paleomagnetism of Archean rocks from northwestern Ontario: Wabigoon gabbro, Wabigoon Subprovince. Canadian Journal of Earth Sciences, **20**, pp. 1805–1817.

DUNLOP, D. J., SCHUTTS, L. D., and HALE, C. J. 1984. Paleomagnetism of Archean rocks from northwestern Ontario: III. Rock magnetism of the Shelley Lake granite, Quetico Subprovince. Canadian Journal of Earth Sciences, **21**. (This issue.)

GEISSMAN, J. W., STRANGWAY, D. W., TASILLO-HIRT, A. M., and JENSEN, L. S. 1982. Paleomagnetism and structural history of the Ghost Range intrusive complex, central Abitibi Belt, Ontario: further evidence for the Late Archean geomagnetic field of North America. Canadian Journal of Earth Sciences, **19**, pp. 2085–2099.

——— 1983. Paleomagnetism of Late Archean metavolcanics and metasediments, Abitibi orogen, Canada: tholeiites of the Kinojevis Group. Canadian Journal of Earth Sciences, **20**, pp. 436–461.

HALLS, H. C. 1976. A least-squares method to find a remanence direction from converging remagnetization circles. Geophysical Journal of the Royal Astronomical Society, **45**, pp. 297–304.

HALLS, H. C., and PESONEN, L. J. 1982. Paleomagnetism of Keweenawan rocks. Geological Society of America, Memoir 156, pp. 173–201.

IRVING, E. 1979. Paleopoles and paleolatitudes of North America and speculations about displaced terrains. Canadian Journal of Earth Sciences, **16**, pp. 669–694.

IRVING, E., and MCGLYNN, J. C. 1981. On the coherence, rotation and palaeolatitude of Laurentia in the Proterozoic. *In* Precambrian plate tectonics. *Edited by* A. Kroner. Elsevier, Amsterdam, The Netherlands, pp. 561–598.

IRVING, E., and NALDRETT, A. J. 1977. Paleomagnetism in Abitibi greenstone belt and Abitibi and Matachewan dikes: evidence of the Archean geomagnetic field. Journal of Geology, **85**, pp. 157–176.

LAPOINTE, P. L., ROY, J. L., and MORRIS, W. A. 1978. What happened to the high latitude paleomagnetic poles? Nature, **273**, pp. 655–657.

LONDRY, J. W., and SYMONS, D. T. A. 1976. Paleomagnetism of Archean rock units and mineralization in the Noranda area, Quebec (abstract). Canadian Association of Physicists Bulletin, **32**, p. 36.

LOWRIE, W., and FULLER, M. 1971. On the alternating field demagnetization characteristics of multidomain thermoremanent magnetization in magnetite. Journal of Geophysical Research, **76**, pp. 6339–6349.

PULLAIAH, G., IRVING, E., BUCHAN, K. L., and DUNLOP, D. J. 1975. Magnetization changes caused by burial and uplift. Earth and Planetary Science Letters, **28**, pp. 133–143.

RIDLER, R. H., and FOSTER, J. H. 1981. Primary paleomagnetic stratigraphy of the Archean Skead pyroclastics, Kirkland Lake area, northeastern Ontario. Geological Survey of Canada, Paper 77-10, 31 p.

SCHUTTS, L. D., and DUNLOP, D. J. 1981. Proterozoic magnetic overprinting of Archean rocks in the Canadian Superior Province. Nature, **291**, pp. 642–645.

SEGUIN, M. K., SHARMA, K. N. M., WOUSSEN, G., ST. SEYMOUR, K., and SYMONS, D. T. A. 1983. Paleomagnetism of Archean volcanics and Proterozoic dikes in La Grande Riviere area of Nouveau-Quebec. Canadian Journal of Earth Sciences, **20**, pp. 339–343.

SYMONS, D. T. A., QUICK, A. W., and STUPAVSKY, M. 1981. Magnetic and paleomagnetic characteristics of the Archean iron formation and host rocks at the Adams mine, Ontario. Ontario Geological Survey, Miscellaneous Paper 98, pp. 293–307.

TASILLO-HIRT, A. M., GEISSMAN, J. W., STRANGWAY, D. W., and JENSEN, L. S. 1982. Paleomagnetism of Late Archean metavolcanics and metasediments, Abitibi orogen, Canada: volcanics of the Blake River Group and sediments and volcanics of the Timiskaming Group. Canadian Journal of Earth Sciences, **19**, pp. 2100–2113.

UENO, H., and IRVING, E. 1976. Paleomagnetism of the Chibougamau Greenstone Belt, Quebec and the effects of Grenvillian post-orogenic uplift. Precambrian Research, **3**, pp. 305–315.

WILSON, R. L., and SMITH, P. J. 1968. The nature of secondary magnetizations in some igneous and baked rocks. Journal of Geomagnetism and Geoelectricity, **20**, pp. 367–380.

ZIJDERVELD, J. D. A. 1967. A.C. demagnetization of rocks: analysis of results. *In* Methods in palaeomagnetism. *Edited by* D. W. Collinson, K. M. Creer, and S. K. Runcorn. Elsevier, Amsterdam, The Netherlands, pp. 254–286.